

Answers to Prelim Exam H2 Physics Paper 2

Qn	Answer	Marks
1(a)	Speed is a physical quantity and can only be defined in terms of other physical quantities. Distance is a physical quantity, but <u>second is a unit for time</u> which cannot be used to define speed.	B1 B1
1(b)	Vertically, from $v = u + gt$ and taking upward as positive, $-u \sin \theta = u \sin \theta - gT$ Time of flight, $T = \frac{2u \sin \theta}{g}$ Horizontally, from $s = ut$ $R = (u \cos \theta)(T)$ $= (u \cos \theta) \left(\frac{2u \sin \theta}{g} \right)$ $= \frac{2u^2 \sin \theta \cos \theta}{g}$	M1 M1 A1 A0
	<i>Marker's comments: It is a "show" type of question and thus important that all steps, including substitution must be explained and shown.</i>	
1(c)	Maximum R occurs when $\sin 2\theta = 1$. Hence $\theta = 45^\circ$	A1
1(d)	$R_0 = \frac{u^2}{g}$ $\frac{45.36 \text{ km}}{1 \text{ h}} = \frac{45.36(1000) \text{ m}}{(60)(60) \text{ s}} = 12.6 \text{ m s}^{-1}$ $g = \frac{u^2}{R_0}$ $= \frac{(12.6)^2}{16.3}$ $= 9.7399 \text{ m s}^{-2}$ $\frac{\Delta g}{g} = 2 \frac{\Delta u}{u} + \frac{\Delta R_0}{R_0}$	C1 A1

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	$\frac{\Delta g}{9.7399} = 2\left(\frac{3}{100}\right) + \frac{4}{100}$ $\Delta g = 0.97$ $= 1 \text{ m s}^{-2} \text{ (to 1 s.f.)}$ Therefore, $g = (10 \pm 1) \text{ m s}^{-2}$	A1 A1
	<i>Marker's comments: Uncertainty should only be left to 1 sig fig. The final answer and its uncertainty should have the same d.p. It is important to convert the speed to m s^{-1} to obtain g in units of m s^{-2}. Also, there were mistakes in understanding the meaning of percentage uncertainty.</i>	